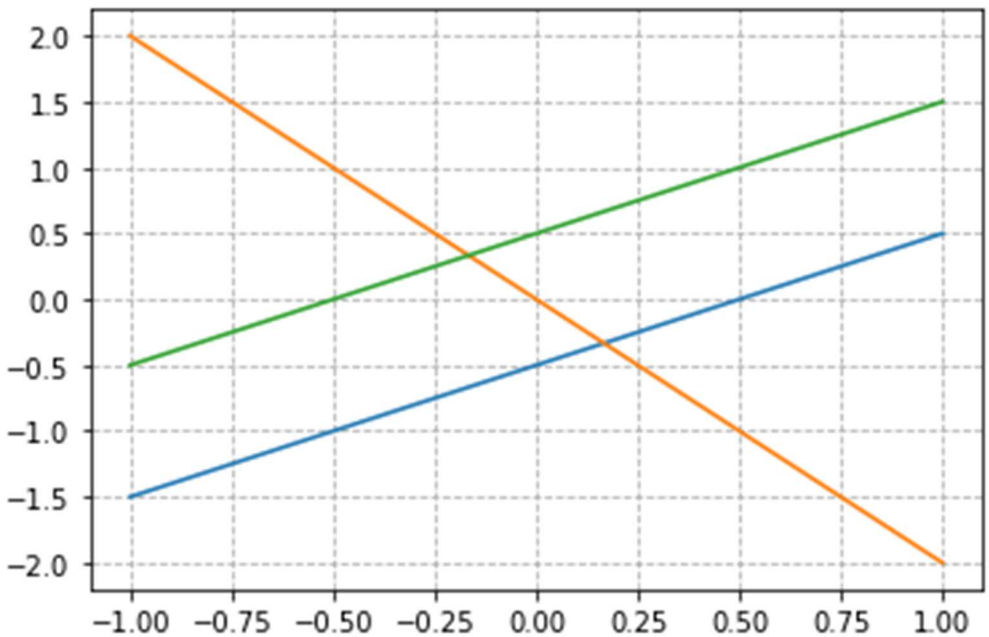
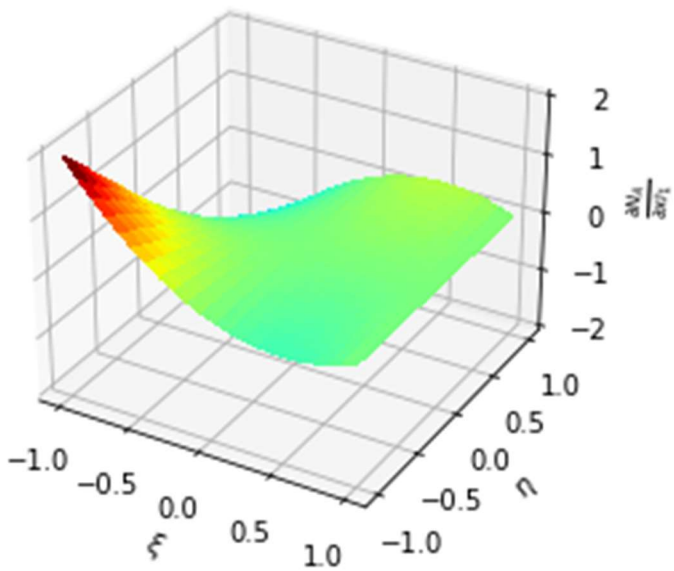
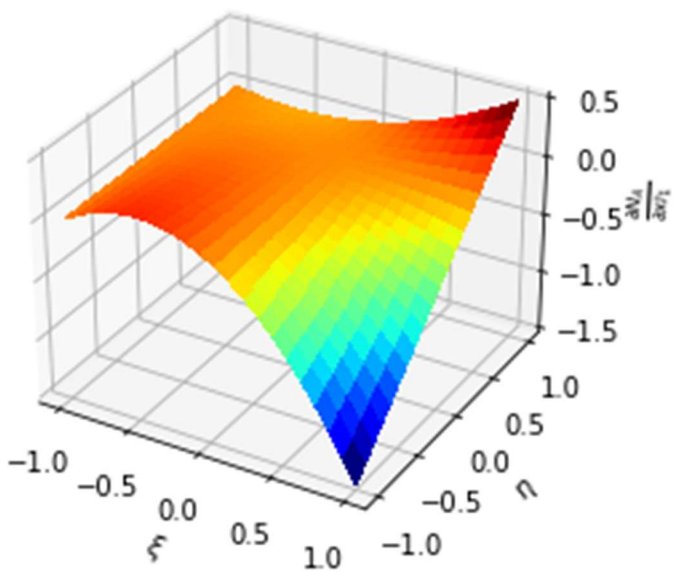
Submit your modified Python code completing each of the above-mentioned functions. Also, submit plots (e.g. in a Word document or as a PDF) for Problems 3 and 4, labeled appropriately.

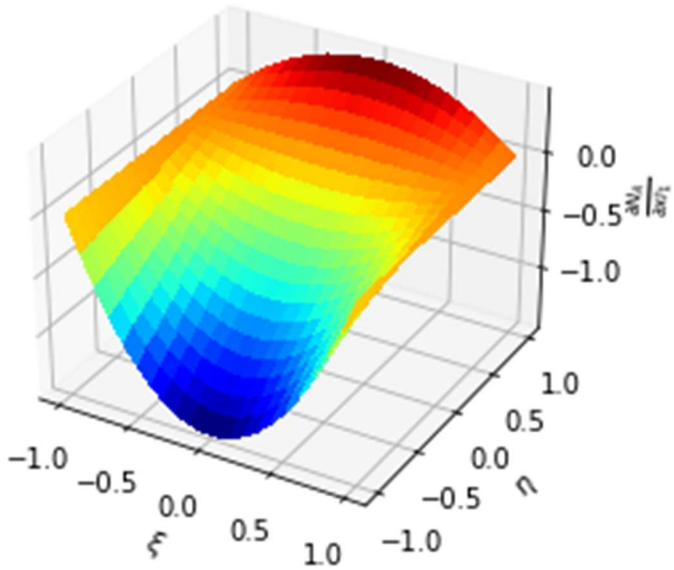
Problem 3: Use the Python function PlotLagrangeBasisDerivatives to plot the derivative of all one dimensional Lagrange basis functions of degree 2 over the interval [-1,1] with interpolation points evenly split in this interval.

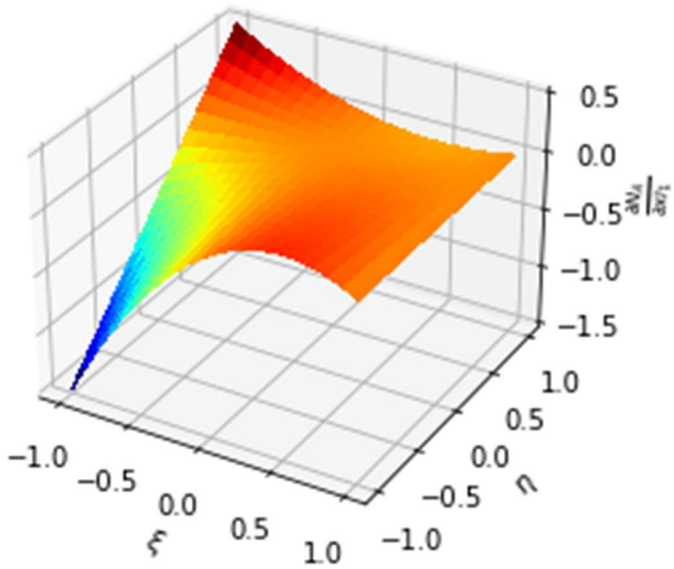


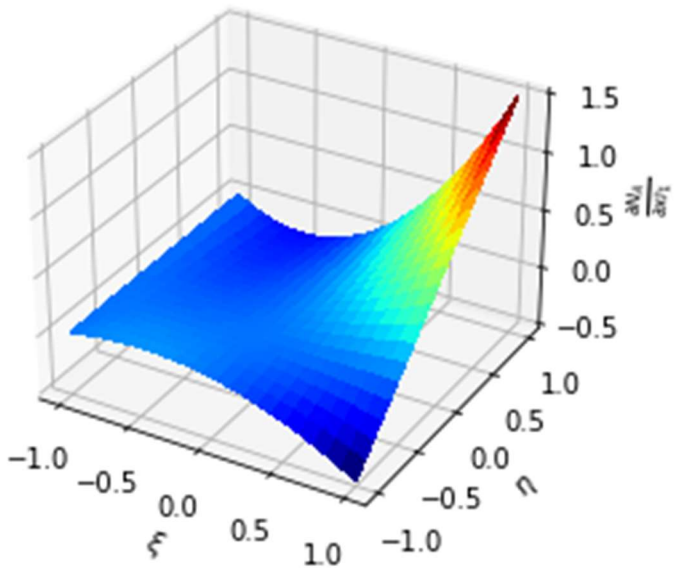
Problem 4: Us the Python function PlotTwoDBasisFunctionParentDomain to plot the partial derivative of all two dimensional Lagrange basis functions of degree 2 in both directions, defined over the square [−1, 1]2 with interpolation points evenly distributed in each interval direction. Plot these for derivatives with respect to ξ and derivatives with respect to η.

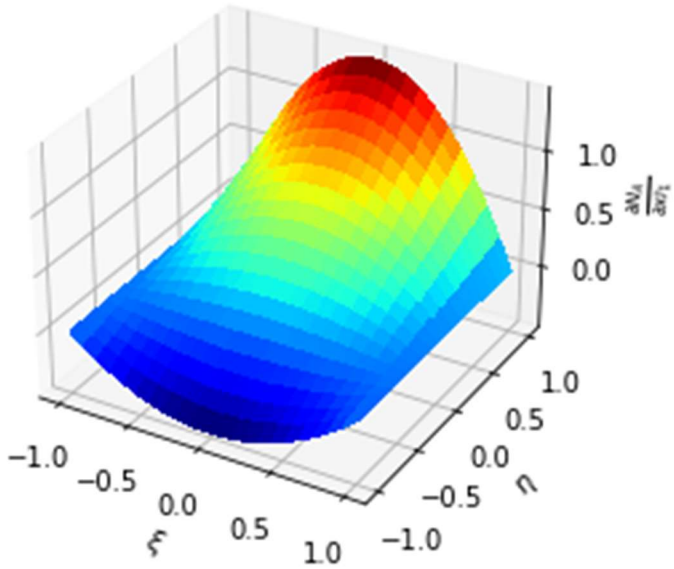


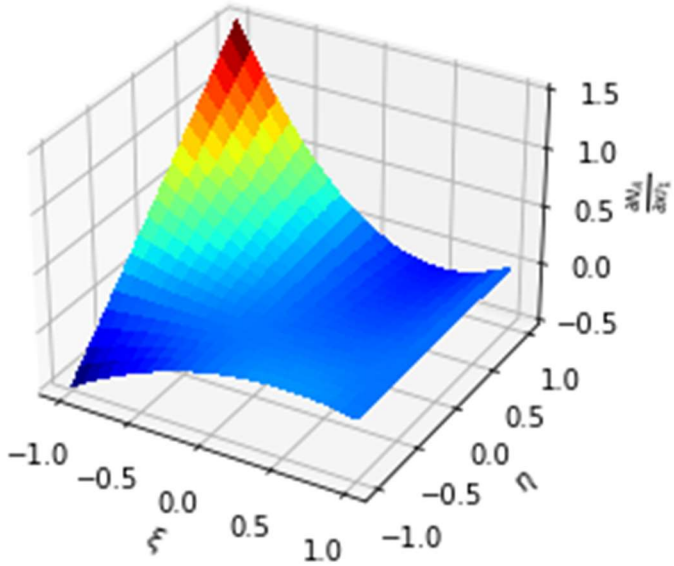


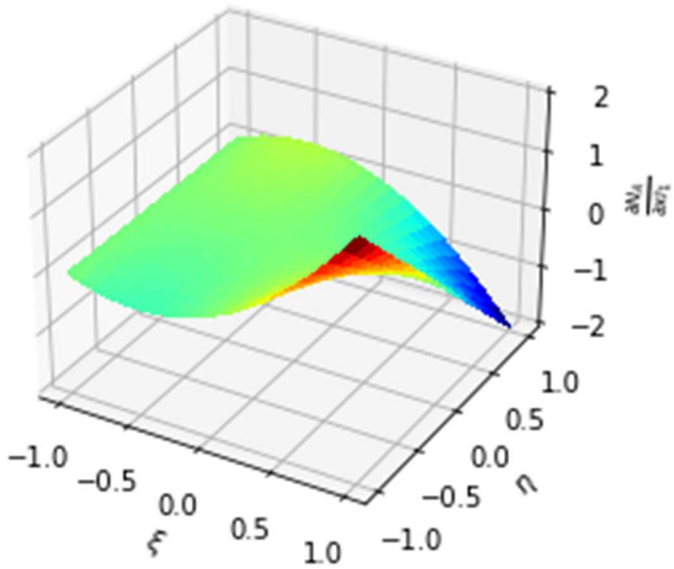

0.0
−0.5
−1.0
$\frac{\partial N_4}{\partial x_1}\Big|_{x_2}$
1.0
0.5
0.0
−0.5
−1.0
η
−1.0
−0.5
0.0
0.5
1.0
ξ

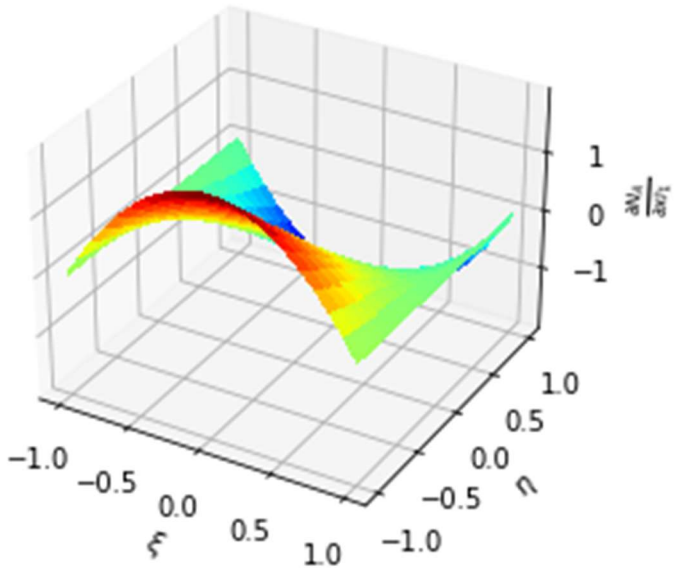

0.5
0.0
−0.5
−1.0
−1.5
$\frac{\partial N_4}{\partial x_1}\Big|_{x_2}$
1.0
0.5
0.0
−0.5
−1.0
η
−1.0
−0.5
0.0
0.5
1.0
ξ


1.5
1.0
0.5
0.0
−0.5
$\frac{\partial N_4}{\partial x_1}$
−1.0
−0.5
0.0
0.5
1.0
ξ
1.0
0.5
0.0
−0.5
−1.0
η


1.0
0.5
0.0
$\frac{\partial N_4}{\partial x_1}$
−1.0
−0.5
0.0
0.5
1.0
ξ
1.0
0.5
0.0
−0.5
−1.0
η


1.5
1.0
0.5
0.0
−0.5
$\frac{\partial N_4}{\partial x_1}$
−1.0
−0.5
0.0
0.5
1.0
ξ
1.0
0.5
0.0
−0.5
−1.0
η


2
1
0
−1
−2
$\frac{\partial N_4}{\partial x_1}$
−1.0
−0.5
0.0
0.5
1.0
ξ
1.0
0.5
0.0
−0.5
−1.0
η


1
0
−1
$\frac{\partial N_a}{\partial \xi_1}$
1.0
0.5
0.0
−0.5
−1.0
η
−1.0
−0.5
0.0
0.5
1.0
ξ